9/25
nuday

CHEMICAL KINETICS.

* Kinetics : study of rate of a rn.
* Rate of a rn :- change in con. of either reactant or pdt with respect to time.

$$A \longrightarrow B$$

$$\text{rate} = \frac{-\Delta[A]}{\Delta t} \text{ or } \frac{+\Delta[B]}{\Delta t}$$

* Average rate $= \frac{\text{total change in con.}}{\text{change in time.}}$

* Instantaneous rate $:= \frac{dx}{dt}$

⇒ Factors affecting Rate of Rn :-

* concentration. (directly proportional)
* Temperature. (directly proportional)
* presence of catalyst. (+ve & -ve) — any substance which alter the rate of rn

Threshold energy :-

The amnt of energy req. for the reactants to become pdt.

Activation Energy :-

The amnt of energy provided to reach threshold energy.

* surface area of reactants.
* radiation ⇒ photochemical rn.
* pressure.

⇒ Rate Law.

$aA + bB \longrightarrow cC + dD$.

rate $\propto [A][B]$

rate $= K[A][B]$

↳ rate constant.

$$\text{Rate} = K[A]^m[B]^n$$

* order of a $Rn \Rightarrow m+n$
- The sum of powers of con. terms in rate law expression.
- experimental.
- can be zero
* molecularity $\Rightarrow a+b$
- The no. of reactant molecules that simultaneously collide with to give product.
- Theoretical
- cannot be zero

⇒ unit of Rate of $R^n \Rightarrow$ $mol\ L^{-1}\ S^{-1}$

⇒ unit of Rate Constant $\Rightarrow$ $mol^{(1-n)}\ L^{(n-1)}\ S^{-1}$

⇒ Zero Order R^n. $(n=0)$ $\Rightarrow mol\ L^{-1}\ S^{-1}$

The con. of rct. doesn't depend on rate.

⇒ First Order R^n. $(n=1)$ $\Rightarrow S^{-1}$

Integrated Rate law for first Order R^n.

	$A \rightarrow$ pdt	$A \longrightarrow$ pdt
at $t=0$;	a	0
t	$(a-x)$	x

$$\frac{dx}{dt} = k_1(a-x)$$

$$dx = k_1(a-x)\cdot dt$$

$$\frac{dx}{a-x} = k_1 \cdot dt$$

$$\int \frac{dx}{a-x} = \int k_1 \cdot dt$$

$$\log^{-}\ln(a-x) = k_1 t + C \quad -(1)$$

At $t=0$; $x=0$; applying the limiting condition.

$$-\ln a = k_1 \cdot 0 + C$$

$$C = -\ln a \quad -(2)$$

Substitute (2) in (1)

$$-\ln(a-x) = k_1 t - \ln a$$

$$\ln\left[\frac{a}{a-x}\right] = k_1 t$$

$$k_1 t = 2.303 \log\left(\frac{a}{a-x}\right)$$

$$\boxed{k_1 = \frac{2.303}{t}\log\left[\frac{a}{a-x}\right]}$$

⇒ **Integrated Rate law for Second Order R^n.**

$$2A \xrightarrow{k_2} Pdt$$

at $t=0$	a	0
t	$(a-x)$	x

$$\frac{dx}{dt} \propto (a-x)^2$$

$$\frac{dx}{dt} = k_2 \cdot (a-x)^2$$

$$\int \frac{1}{x^2} = -\frac{1}{x}$$

$$\int \frac{dx}{(a-x)^2} = \int k_2 \cdot dt$$

$$\frac{-1}{a-x} \times -1 = k_2 \cdot t + C$$

$$\frac{1}{a-x} = k_2 \cdot t + C \quad -(1)$$

Applying limiting condn : at $t=0$; $x=0$;

$$\frac{1}{a} = k_2 \cdot 0 + C$$

$$C = \frac{1}{a} \quad -(2)$$

(2) in (1)

$$\frac{1}{a-x} = k_2 t + \frac{1}{a}$$

$$k_2 t = \frac{1}{a-x} - \frac{1}{a}$$

$$\boxed{k_2 = \frac{1}{t}\left[\frac{x}{a(a-x)}\right]}$$

⇒ Integrated Rate law for Third-Order Rⁿ.

$$3A \longrightarrow \text{pdt}$$

at $t=0$	a	0
t	$a-x$	x

$$\frac{dx}{dt} \propto (a-x)^3$$

$$\frac{dx}{dt} = k_3 (a-x)^3$$

$$\int \frac{dx}{(a-x)^3} = \int k_3 \cdot dt$$

$$\frac{-1}{2(a-x)^2} \times -1 = k_3 t + C$$

$$\frac{1}{2(a-x)^2} = k_3 t + C$$

when $x=0$; $t=0$

$$\frac{1}{2a^2} = k_3 \times 0 + C$$

$$C = \frac{1}{2a^2}$$

$$\frac{1}{2(a-x)^2} = k_3 t + \frac{1}{2a^2}$$

$$k_3 t = \frac{1}{2a^2(2(a-x)^2)}$$

$$k_3 = \frac{1}{2t}\left[\left(\frac{1}{a^2 x}\right)^2 - \frac{1}{a^2}\right]$$

$$x^{-3}$$

$$\frac{x^{n+1}}{n+1}$$

$$\frac{x^{-3+1}}{-3+1}$$

$$\frac{x^{-2}}{-2}$$

$$\frac{-1}{2x^2}$$

→ Integrated Rate law for Zero Order Rxn.

$$\frac{dx}{dt} = k_0 (a-x)^0$$

$$\frac{dx}{dt} = k_0$$

$$\int dx = \int k_0 \cdot dt$$

$$x = k_0 t + C$$

when $x=0$, $t=0$;

$$C = 0$$

$$x = k_0 t$$

$$k_0 = \left[\frac{x}{t}\right]$$

→ Half life ($t_{1/2}$)

The time taken to complete half of the con- of reactants of a rxn.

$$k_1 = \frac{2.303}{t} \log\left[\frac{a}{a-x}\right]$$

when $t \to t_{1/2}$; $x = \frac{a}{2}$

$$k_1 = \frac{2.303}{t_{1/2}} \log\left[a/a_{/2}\right]$$

$$t_{1/2} = \frac{2.303}{k_1} \log 2$$

(0.3010)

$$t_{1/2} = \frac{0.693}{k_1}$$

* $$k_2 = \frac{1}{t}\left[\frac{x}{a(a-x)}\right]$$

when $t \to t_{1/2}$; $x = \frac{a}{2}$

$$k_2 = \frac{1}{t_{1/2}}\left[\frac{a/2}{a(a-a/2)}\right]$$

$$k_2 = \frac{1}{t_{1/2}}\left[\frac{a/2}{a \times a/2}\right]$$

$$k_2 = \frac{1}{t_{1/2}}\left[\frac{1}{a}\right]$$

$$t_{1/2} = \frac{1}{k_2 a}$$

* $$k_3 = \frac{1}{2t}\left[\frac{1}{(a-x)^2} - \frac{1}{a^2}\right]$$

when $t_{1/2} \to t_{1/2}$ $\quad x = \frac{a}{2}$

$$k_3 = \frac{1}{2t_{1/2}}\left[\frac{1}{\left(a-\frac{a}{2}\right)^2} - \frac{1}{(a/2)^2}\right]$$

$$= \frac{1}{2t_{1/2}}\left[\frac{1}{(a/2)^2} - \frac{1}{a^2}\right]$$

$$t_{1/2} = \frac{1}{2k_3}\left[\frac{\frac{a^2}{4} - a^2}{\frac{a^2}{4} \times a^2}\right] \qquad t_{1/2} = \frac{1}{2k_3}\left[\frac{4}{a^2} - \frac{1}{a^2}\right]$$

$$t_{1/2} = \frac{3}{2k_3a^2}$$

* $k_0 = \frac{x}{t}$

$$t_{1/2} = \frac{a}{2k_0}$$

⇒

order	$t_{1/2}$
0	$\propto a$
1	independedt
2	$\propto \frac{1}{a}$
3	$\propto \frac{1}{a^2}$

$$t_{1/2} \propto \frac{1}{a^{(n-1)}}$$

The decomposition of a substance follows first order kinetics. $t_{1/2}$ is 35 mins. What is the rate constant.

$k_1 =$

$$t_{1/2} = \frac{0.693}{k_1}$$

$$\therefore k_1 = \frac{0.693}{35 \times 60} = 0.0198 \text{ min}$$

Q: The thermal decomposition of a compound is of first order. If 5% of it is decomposed in 120 mins, how long will it take for 90% of the compound to decompose.

$$t_{1/2} = 120 \text{ min}$$

$$K_1 = \frac{0.693}{t_{1/2}} = \frac{0.693}{120 \times 60} = 5.77 \times 10^{-3}$$

$$t = \frac{2.303}{K_1} \log \left[\frac{a}{a-x} \right]$$

$$t = \frac{2.303}{(5.77 \times 10^{-3})} \log \left[\frac{100}{100-90} \right]$$

$$t = \frac{2.303}{5.77 \times 10^{-3}} \log 10$$

$$= 399.1 \text{ min}.$$

Q: Calculate two third ($\frac{2}{3}^{rd}$) life of a first order $\mathcal{R}^n$ having rate constant $K = 5.48 \times 10^{-14} \, s^{-1}$

$t \rightarrow t_{2/3}$

$x \rightarrow \frac{2a}{3}$

$$K_1 = \frac{2.303}{t} \log \left[\frac{a}{a-x} \right]$$

$$t_{2/3} = \frac{2.303}{K_1} \log \left[\frac{a}{a-\left(\frac{2a}{3}\right)} \right]$$

$$= \frac{2.303}{K_1} \log \left[\frac{3a}{a/3} \right]$$

$$= \frac{2.303}{5.48 \times 10^{-14}} \times \log 3$$

$$= \frac{2.303}{5.48 \times 10^{-14}} \times 0.477$$

$$= 2.004 \times 10^{13}$$

$$= 2 \times 10^{13} \text{ sec.}$$

1:-

P_0 (mm of Hg)	:	98	230	296
$t_{1/2}$ (s)	:	86	85	86

Since initial concentration (P_0) & $t_{1/2}$ are independent.
It is of First Order.

2:- Identify the order of the n^n, for which the rate constants are 5.5×10^{-2} mol $L^{-1} s^{-1}$, 3.1×10^{-2} $mol^{-1} L^1 s^{-1}$, $2.5 \times 10^{-2} s^{-1}$, 1.5×10^{-2} $mol^{-2} L^2 s^{-1}$, 1.5×10^{-2} $atm^{-1} s^{-1}$

$mol^{1-n} L^{n-1} s^{-1}$

① 5.5×10^{-2} mol $L^{-1} s^{-1}$ $\therefore n=0 \Rightarrow$ Zero Order.

② 3.1×10^{-2} $mol^{-1} L^1 s^{-1}$ $\therefore n=2 \Rightarrow 2^{nd}$ order.

③ $2.5 \times 10^{-2} s^{-1}$ $\therefore n=1 \Rightarrow 1^{st}$ order.

④ 1.5×10^{-2} $mol^{-2} L^2 s^{-1}$ $\therefore n=3 \Rightarrow 3^{rd}$ order.

⑤ 1.5×10^{-2} $atm^{-1} s^{-1}$ $\therefore n=2 \Rightarrow 2^{nd}$ order.

$atm \approx mol/L$

$\therefore atm^{-1} = mol^{-1} L$